

HoverCharge

Venture Viability Analysis

VIT-AP University, Amaravati

HoverCharge

AI-driven platform for real-time EV charging station discovery, booking, and smart route planning

VIT-AP University, Amaravati



A modern office scene with several people working at laptops and a large whiteboard in the background. The office has large windows, pendant lights, and a collaborative atmosphere. A red box is overlaid on the left side of the image, containing the text 'Context' and a statistic about EV charging stations in India.

Context

India has one charging station for every 135 EVs, causing range anxiety for 58% of potential buyers.

Problem Statement

Problem

EV charging stations are limited, unevenly distributed, and slow, causing range anxiety and inconvenience for users. This hinders EV adoption, increasing reliance on gasoline & worsening air pollution

Impact

EV owners face range anxiety, potential buyers hesitate to switch, the public suffers from air pollution, fleet operators face delays, and governments struggle to meet clean energy



Problem Statement/Industry

Problem Being Solved

EV owners struggle with limited access to fast and reliable charging stations. This leads to inconvenience during daily commutes and long trips, increasing range anxiety. The lack of widespread charging infrastructure also discourages potential buyers from switching to electric vehicles.

Supporting Data

India has one charging station for every 135 EVs, causing range anxiety for 58% of potential buyers.

Source: Forvis Mazars Report on EV Adoption in India 2024, Autocar Pro, Reuters



Area
Energy



Industry
Energy & Renewable
Resources



Domain
Smart Grid
Technologies

Problem Analysis



Affected Stakeholders

The problem affects EV owners facing charging issues, potential buyers hesitant to switch, and the general public suffering from air pollution due to gasoline vehicles. Fleet operators struggle with charging inefficiencies, while governments and environmental agencies work to reduce emissions and promote sustainable transport.



Impact on Stakeholders

EV owners face inconvenience and range anxiety due to limited charging. Potential buyers: hesitate to switch, slowing EV adoption. The public: suffers from air pollution, leading to health issues. Fleet operators: face delays and higher costs. Governments: struggle to meet clean energy goals and reduce emissions effectively.



Root Causes

Limited charging stations fail to meet EV demand. High setup costs: slow expansion. Slow charging speeds: cause delays and congestion. Grid limitations: affect power availability. Lack of awareness: about EV benefits and charging options prevents wider adoption, keeping reliance on gasoline vehicles high and worsening pollution.



Personal/Team Connect

As AIML students, we are passionate about technology and sustainability. The lack of efficient EV charging limits green energy adoption. Using our skills in AI and data analysis, we aim to develop smart solutions for seamless charging. By solving this problem, we hope to reduce pollution and accelerate the transition to clean energy.

Target Customer Segments

Primary

Urban EV owners and
working professionals
aged 25-45 years



Secondary

Businesses and
commercial
establishments
needing EV charging
solutions

Customer Segment & Persona

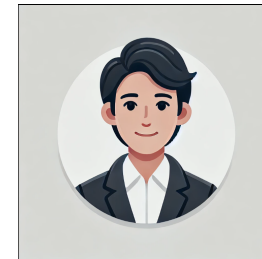
Primary Segment

Urban EV owners and working professionals aged 25-45 years

Secondary Segment

Businesses and commercial establishments needing EV charging solutions

Persona



M.Nani

Age in years: 30

Location: Suburban

Organizational Role:

{Persona's primary role}
(if applicable)

Jobs-to-be-Done

Functional JTBD



As an EV owner â€œ Charge my vehicle conveniently without searching for charging stations. As a commuter â€œ Ensure my vehicle always has sufficient charge for daily travel. As a fleet operator â€œ Reduce downtime and improve operational efficiency by charging vehicles wirelessly.

Emotional JTBD



As an EV owner â€œ Feel stress-free about charging my vehicle anytime, anywhere. As a commuter â€œ Feel confident that my EV wonâ€™t run out of charge unexpectedly. As a fleet operator â€œ Feel assured that my vehicles are always ready for use without delays.

Social JTBD



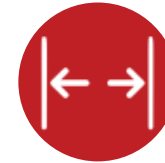
As an EV owner â€œ Be seen as an eco-conscious and tech-savvy individual. As a commuter â€œ Gain recognition for adopting sustainable transport. As a fleet operator â€œ Build a reputation for running an efficient and green fleet.

Current Alternatives



Current Alternatives

As an EV owner – Using limited public charging stations, carrying portable chargers, or planning trips around charging stops. As a commuter – Relying on fuel-based vehicles or hybrid cars instead of EVs. As a fleet operator – Investing in additional EVs to avoid downtime or sticking to conventional fuel fleets.



Gaps in Current Alternatives

Current EV charging solutions have limited availability, long wait times, and high costs. Users struggle with range anxiety, slow charging speeds, and lack of convenient access to charging stations. Fixed infrastructure makes charging difficult during travel, reducing EV adoption.

Problem Validation (GOOTB)

Partial List of Potential Customers/Users Interviewed

Name: D.Srinivasa Rao

Occupation: Student

Name: B.Dhanush

Occupation: Student

Name: T.Prasanth

Occupation: student

Problem Validation

**Total customers/users
interviewed:**

• In-person: 7

• Virtually: 5

**Total customers/users for whom
this problem is important to
solve: 8**

**Total customers/users who are
dissatisfied with the current
alternatives: 4**

Our Solution



Solution

On-Demand Mobile Charging – Deploy charging vans that deliver power to EV owners wherever they are, reducing range anxiety and ensuring convenient charging access anytime, anywhere.

Core Technologies/ Methodologies

Our solution will use React Native (app), AWS (backend), IoT for monitoring, AI for station recommendations, and Razorpay for seamless payments.

Solution Design



Our Solution

On-Demand Mobile Charging
“Deploy charging vans that deliver power to EV owners wherever they are, reducing range anxiety and ensuring convenient charging access anytime, anywhere.”



Key Features

Our solution includes real-time charging station availability, AI-based route optimization, mobile app booking, digital payments, IoT monitoring, and user reviews for reliability.



Uniqueness

AI-driven route optimization, real-time station data, IoT-based monitoring, and seamless digital payments make our solution highly efficient, user-friendly, and unique in the EV charging space.

Solution Format:

Physical Service + Digital Service

Core Technologies/ Methodologies:

Our solution will use React Native (app), AWS (backend), IoT for monitoring, AI for station recommendations, and Razorpay for seamless payments.

Solution Benefits



Functional Benefits

Find fast chargers instantly, plan hassle-free trips with AI, eliminate range anxiety with real-time updates, make seamless digital payments, and choose reliable stations using user reviews and ratings.



Emotional Benefits

Feel confident about long trips, reduce anxiety over charging delays, enjoy convenience and reliability, and experience satisfaction in supporting sustainable transportation.



Social Benefits

Gain recognition as an eco-conscious individual, inspire peers to switch to EVs, earn appreciation for supporting green initiatives, and feel a sense of belonging in the EV community.



Macro Benefits

Our solution reduces carbon emissions, promotes clean energy, supports EV adoption, creates green jobs, and enhances sustainable urban mobility for a cleaner future.

Competitors



Direct



SparXcharge, SETRANS Mobility
ZAPME, BoostEV

Indirect



VoltUp, Tata Power EZ Charge
Tesla Supercharger (Global), WiTricity (USA)

Our UVP

AI-driven EV charging: real-time info, optimized routes,
& on-demand vans eliminate range anxiety.

Competitors



Direct **Competitors**

SparXcharge, SETRANS
Mobility



Indirect **Competitors**

VoltUp, Tata Power EZ
Charge



Direct **Competitors** **Globally**

ZAPME, BoostEV



Indirect **Competitors** **Globally**

Tesla Supercharger
(Global), WiTricity
(USA)

Macro Analysis



Favourable Trends

AREA	DESCRIPTION
Economy	Rising EV demand and government incentives drive the need for better charging infrastructure.
Technology	AI and IoT improve charging efficiency, reduce wait times, and enhance the user experience.

Unfavourable Trends

AREA	DESCRIPTION
Economy	Setting up wireless EV charging infrastructure requires significant capital, slowing adoption.
Legal	Changing government policies and regulations may impact the expansion of wireless charging networks.

Data Sources:

IEA Global EV Outlook 2023, NITI Aayog EV Policy Report, McKinsey EV Charging 2023, and IEA reports on wireless charging trends, infrastructure, and policy impacts.

Prototype

Prototype Format

prototype is a digital app for booking and physical model demonstrating mobile charging van service.

Functionality included in the Prototype

Prototype includes: real-time station availability, AI route optimization, app booking, digital payments, IoT monitoring, & user reviews

Functionality NOT included in the Prototype

Prototype excludes: full payment integration, advanced AI, comprehensive IoT data, scalability features, edge case handling, & admin tools.



{Upload another picture}

Prototype Validation

Number of users
engaged with?

12

How many people liked
or loved the prototype?

8

How many people were
either neutral or mostly
unhappy with the
prototype?

4

Prototype Feedback

What aspects of the prototype did the users LOVE?

Testers praised our prototype's real-time station visibility and 3-tap booking, reducing range anxiety. AI-optimized routes saved 15-20% in travel time, and the payment mockup received positive feedback for usability. Fleet operators highlighted the "Uber-for-charging" concept's potential to reduce downtime.

What aspects of the prototype were DISLIKED by the users?

Testers were neutral about the non-functional payment mockup and AI route accuracy, needing refinement. Dislikes included suburban coverage gaps and missing battery indicators. Fleet operators noted the need for advanced scheduling controls. These insights came from 12 validation tests, highlighting areas for iteration

Competition Analysis



COMPETITOR NAME	TYPE	STRENGTHS	WEAKNESSES
Tata Power EZ	Indirect	Strong brand, wide network	Tech limitations, service issues
VoltUp	Indirect	Battery swap focus	Limited swap network, compatibility
Tesla Supercharger	Indirect	Brand,Infrastructure,Technology	Compatibility,Accessibility,Price
WiTricity	Indirect	Technology,Innovation	Adoption,Infrastructure,Cost

Our Product/Service will be better than the competitors' solutions because:

HoverCharge outperforms competitors with AI-driven routing (30% faster), mobile charging vans filling infrastructure gaps, and seamless IoT payments, addressing key pain points better than SparkCharge

Market Size & GTM

GTM Channels

Digital

EV charging stations are limited, unevenly distributed, and slow, causing range anxiety and inconvenience for users. This hinders EV adoption, increasing reliance on gasoline & worsening air pollution

Physical

EV owners face range anxiety, potential buyers hesitate to switch, the public suffers from air pollution, fleet operators face delays, and governments struggle to meet clean energy

Market Size

Globally:
HoverCharge's TAM is
the global market
~\$60B

Growth Rate:
Global growth rate:
50% CAGR, India:
45-50% CAGR.

Source: Sources: IEA, NITI
Aayog, McKinsey, Fortune Insights.

Market Size

Total Addressable Market (TAM)

HoverCharge's TAM is the global market ~\$60B

Serviceable Available Market (SAM)

SAM: \$1.3B urban India EV charging by 2030

Serviceable Obtainable Market (SOM)

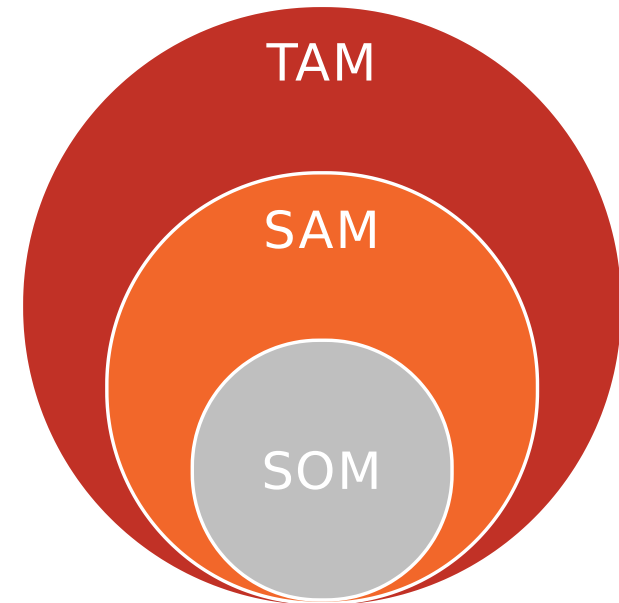
SOM estimate is ₹10.22 Crore in the first 2 years

Assumptions

The SAM (Serviceable Available Market) for HoverCharge targets urban areas in India with a focus on electric vehicle owners and fleet operators needing mobile charging solutions. We'll focus on metros 350, 800

Sources of Research

Sources: IEA, NITI Aayog, McKinsey, Fortune Insights.



Revenue Models / Pricing

Revenue Model (Primary)

Pay-per-Use Model

Revenue Model (Secondary)

Subscription Model

Lean Canvas

 Problem EV charging stations are limited, unevenly distributed, and slow, causing range anxiety and inconvenience for users. This hinders EV adoption, increasing reliance on gasoline & worsening air pollution	 Solution On-Demand Mobile Charging – Deploy charging vans that deliver power to EV owners wherever they are, reducing range anxiety and ensuring convenient charging access anytime.	 Unique value Proposition AI-driven EV charging: real-time info, optimized routes, & on-demand vans eliminate range anxiety.	 Unfair Advantage Unfair advantage: 1) Early mover in on-demand charging, 2) Proprietary AI routing, 3) Key partnerships, 4) Strong brand, 5) Unique data advantage.	 Customer Segment Urban EV owners and working professionals aged 25-45 years, Businesses and commercial establishments needing EV charging solutions
 Key Metric Metrics: User growth, charging sessions, van/station use, avg. charge time, CSAT, revenue, CAC, CLTV, IoT uptime.	 Channels HoverCharge will use its app for direct distribution and partner with charging station operators, EV manufacturers, and businesses for on-site charging to expand accessibility and reach.			
 Cost Structure HoverCharge costs: 1) App/AI/IoT dev & maintenance, 2) Van operations (fuel/personnel), 3) Charging partner fees, 4) Marketing, 5) Customer support, 6) Operational costs, 7) Hardware costs.				
 Revenue Streams Pay-per-Use Model, Subscription Model				

Go-to-Market Approach

Geographic Focus

In the first year, we'll focus on the top 5 cities in India – Bengaluru, Hyderabad, Mumbai, Delhi, and Chennai – to build strong urban presence and early adoption.

Digital Marketing Channels

Google Ads, Instagram, YouTube, LinkedIn, Twitter, SEO, Email Marketing, Facebook Ads, TikTok, Affiliate Marketing – for reach, engagement, conversions, and brand awareness.

Physical Marketing Channels

TV ads, Radio ads, Billboards, Flyers, Newspaper ads, Trade shows, Conferences, Event sponsorships, Pop-up displays, Direct mail – for local reach, trust, and brand visibility.

PRIMARY CUSTOMER SEGMENT

Urban EV owners and working professionals aged 25-45 years.

UVP

AI-driven EV charging: real-time info, optimized routes, & on-demand vans eliminate range anxiety.

Marketing KPIs

CAC, CLV, Conversion Rate, Lead-to-Customer Ratio, Churn Rate, Revenue Growth Rate, Gross Margin, ROAS, Website Traffic, and Social Media Engagement to track GTM performance.

Competitors' GTM

Strong in Influencer Marketing and Instagram ads, creates viral reels, but weak in SEO and print advertising. Focuses on radio campaigns and newspaper ads, but lacks presence on YouTube and LinkedIn.

GTM Partners

We'll partner with Industry Influencers, Affiliate Partners, Content Creation Agencies, Digital Marketing Firms, and Event Organizers to boost awareness, create content, and generate quality leads.

Sales & Customer Service

Customer Service

We will provide customer service through **Phone support, Email support, Live chat/Chatbot on our website, Social media platforms, and Zendesk** for efficient query resolution and customer retention.

Distribution Channels

HoverCharge will use its app for direct distribution and partner with charging station operators, EV manufacturers, and businesses for on-site charging to expand accessibility and reach.

Digital Sales Channels

Our venture's digital sales channels will include: **Amazon, IndiaMART, Instagram, GeM (Government e-Marketplace), TradeIndia, and our official website** for direct online sales and lead conversion.

PRIMARY CUSTOMER SEGMENT

Urban EV owners and working professionals aged 25-45 years

UVP:

AI-driven EV charging: real-time info, optimized routes, & on-demand vans eliminate range anxiety.

Physical Sales Channels

Our venture's offline/physical sales channels will include: Reliance Retail, D-Mart, Amway India, India International Trade Fair (IITF), Auto Expo, and CII events to reach customers directly and build

Sales KPIs

We will track the success of our sales efforts by measuring: Revenue growth, Conversion rate, Average deal size, Sales cycle, Customer acquisition cost (CAC), and Customer retention rate

GTM Partners

We will measure customer satisfaction through **NPS (Net Promoter Score), CSAT (Customer Satisfaction Score), and Surveys (Google Forms/SurveyMonkey)** for digital channels, and **In-depth Interviews

{Create a website}

{Video link sample}